Agentic Capabilities Orchestration

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Ontology-guided agent orchestration



THE PROBLEM

Which skills and tools?
In what order? Why this plan?

OUR APPROACH

We are developing an ontology that connects user needs, capabilities and dependencies.

Explainable plan selection and traceable execution

Team AgenticCapabilities

Personalized supplement recommendations

CURRENT SCOPE

Supplement Advisor MCP	Candidate supplements
PubMed MCP	Research evidence
OpenFDA MCP	Safety information
iHerb-CLI	Product search

STAKEHOLDERS

LLM users, agent developers and external service providers

MCP / CLI only. RAG and GraphRAG are outside scope.



Query: “Compare magnesium glycinate and magnesium citrate for supporting sleep, and summarize the scientific evidence for each.”

Constraints: minimize execution time, safety screening

Suppose the chosen orchestration plan is:

1a. Verify evidence

PubMed: search_articles(magnesium glycinate for sleep) => PMIDs
PubMed: download_articles_batch(PMIDs) => article abstracts
summarize_evidence()

1b. Screen interactions

OpenFDA: search_drug_adverse_events(magnesium glycinate)
OpenFDA: search_drug_labels(magnesium glycinate)
OpenFDA: search_drug_recalls(magnesium glycinate)

2a. Verify evidence

PubMed: search_articles(magnesium citrate for sleep) => PMIDs
PubMed: download_articles_batch(PMIDs) => article abstracts
summarize_evidence()

2b. Screen interactions

OpenFDA: search_drug_adverse_events(magnesium citrate)
OpenFDA: search_drug_labels(magnesium citrate)
OpenFDA: search_drug_recalls(magnesium citrate)

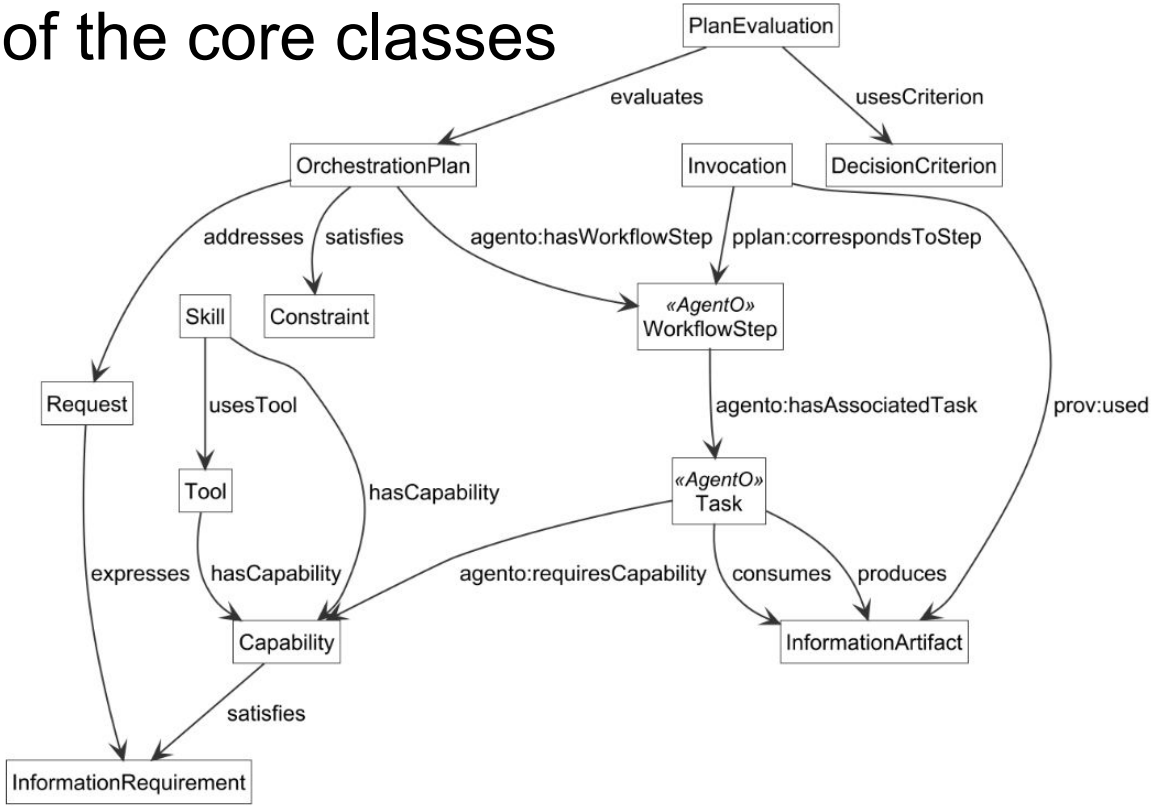
3. Synthesize evidence

CQ: Why were the agent capabilities for a given orchestration invoked in this order?

Answer

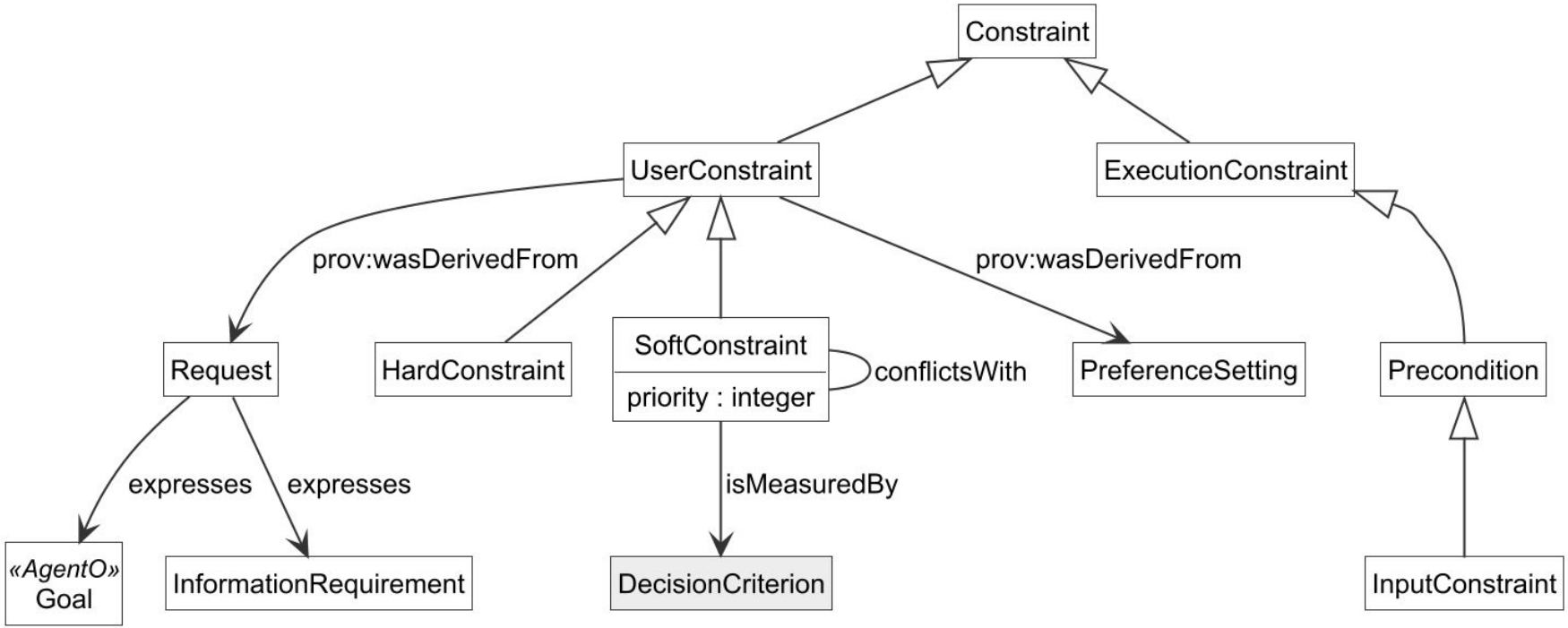
- Hard constraint: **safety screening**. The **Screen interactions skill** satisfies that constraint, it must be included in the orchestration plan
- Soft constraint: minimize execution time. The **Verify evidence skill** and **Screen interactions skill** require the same dependencies (**candidate supplement**). They can be invoked in parallel to minimize execution cost

Overview of the core classes

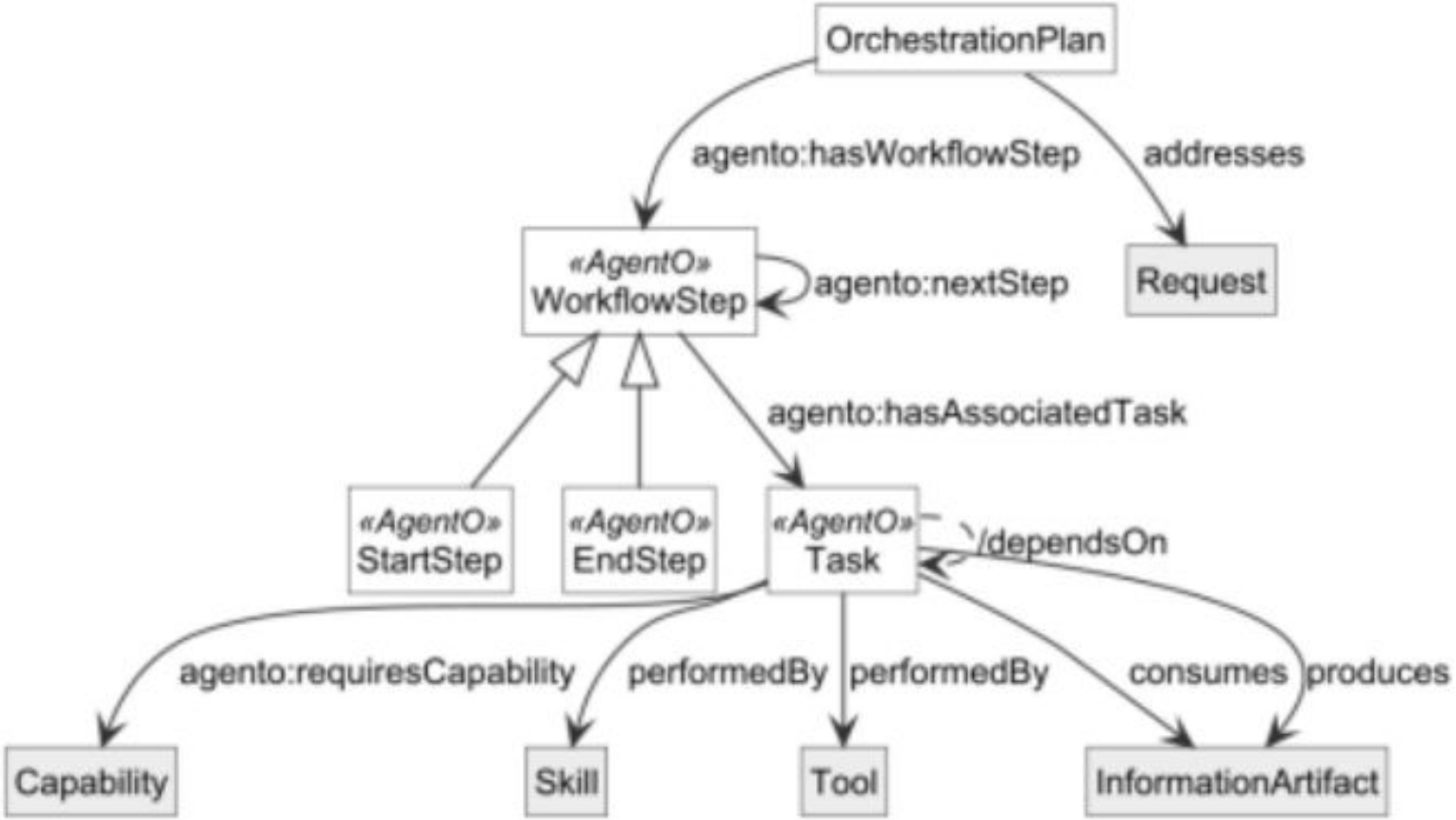


Request → Requirements → Capabilities → Candidate Plans → Evaluation → Selected Plan → Execution

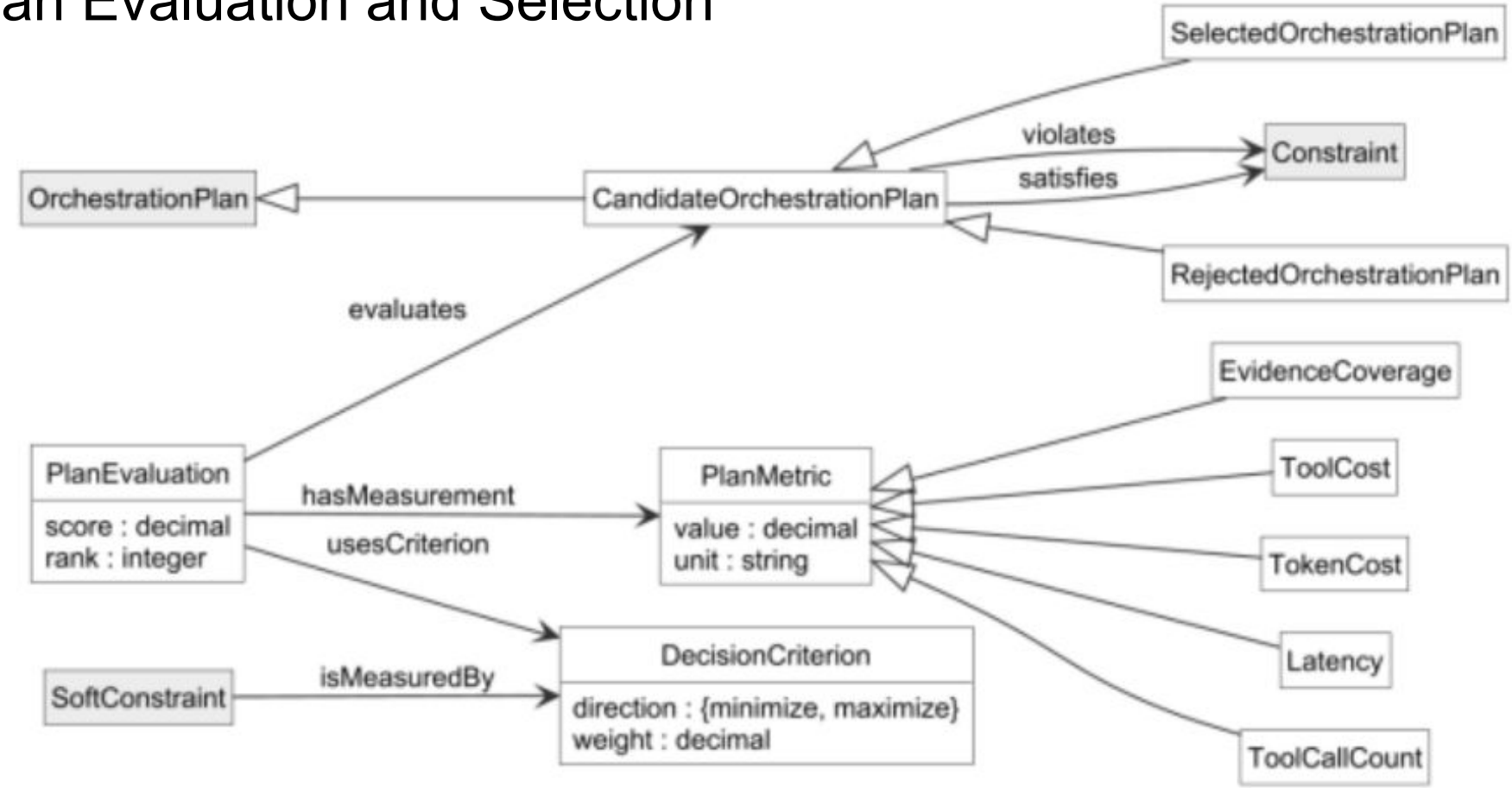
Requests, requirements and constraints



Plan Structure and Task Dependencies



Plan Evaluation and Selection



GenAI Use 1/2 — Claude drafted our conceptual model

Claude Fable 5.1 (Anthropic), Claude Code desktop app · Sept 30 – Oct 1 · output: PlantUML source

1 CONTEXT WE GAVE IT

The prompt only set the goal: draft our A5 conceptual model. The content came from six files Claude read directly.

A5 handout and rubric
Targets: ≥30 classes, ≥15 properties

Week 5 lecture
True is-a, naming, n-ary pattern

A4 use case
Scenarios and the 9 CQs to answer

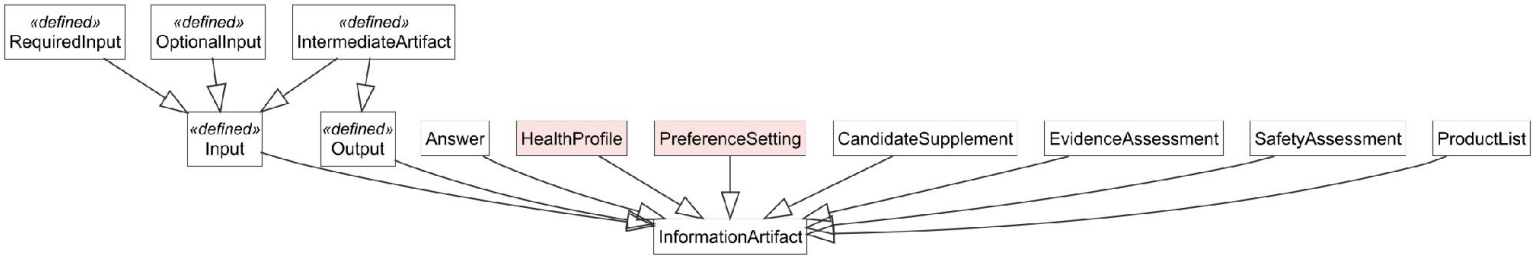
52-term curated list
Names and definitions to trace to

AgentO ontology file
Terms to reuse, domains and ranges

Danielle's A4 comments
Conflicting criteria → hard vs. soft

2 IT RETURNED

Within an hour: 8 UML class diagrams in PlantUML, each tied to the CQs, and a term-list traceability workbook.



Snippet: Figure 8 as first written (Sept 30, 1:41 am), rendered from its PlantUML source. The two red classes were removed in v1.1.

3 WE KEPT, WE CHANGED

KEPT FROM THE DRAFT

- Hard vs. soft constraints (conflicting criteria)
- Skill and Tool kept apart from Capability
- PlanEvaluation as an n-ary class

CHANGED BY THE TEAM (V1.1, OCT 1)

- HealthProfile removed (dropped from the use case)
- PreferenceSetting out of InformationArtifact
- CQs renumbered to the A5 order

GenAI Use 2/2 — How we checked it, and the other uses

Checked against the source files · ChatGPT gave a worked example for CQ 6

HOW WE CHECKED THE CLAUDE DRAFT

- Rendered and inspected every diagram
- Counted from the source: 64 classes, 34 object properties, 4 is-a levels
- Checked reused AgentO properties against the AgentO file:

> *"Three AgentO properties are not reused: hasGoal, hasCapability and performedBy. Their AgentO definitions would make a reasoner conclude a request is an agent and a skill is a tool …"*
> Claude, Sept 30, 1:53 am

- Checked every is-a link against our term-list definitions: two A4 errors removed
- Walked all 9 CQs through the diagrams; team review gave v1.1

CHATGPT DESKTOP: A WORKED EXAMPLE FOR CQ 6

Context: two candidate plans written out as their tool calls, and two decision criteria (Evidence Coverage, Minimize External Tool Calls).
Goal: score each plan from 0 to 1.

Plan	Evidence coverage	Min. tool calls
Plan A (10 tool calls)	1.00	0.00
Plan B (4 tool calls)	0.70	1.00

Each score is one PlanEvaluation in Figure 6 (plan, criterion, score). Chat link

SMALLER USES THIS WEEK

- **ChatGPT desktop:** revised the structural diagram so the answer returns to the user. Chat link
- **ChatGPT desktop:** preferred-label search; none fit, we kept ours
- **GPT-6 Astra:** compared diagram terms with the term list for gaps

WHAT WE WOULD TELL ANOTHER TEAM

Give it the files, not a summary
The files exposed 2 is-a errors and 3 unsafe AgentO reuses

Ask for open questions first
It drafted from the A4 use case, so HealthProfile came out later

Ask for a 20–30 class core first
64 classes at once was too many to review in one sitting